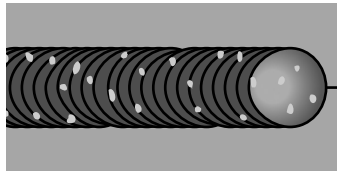


Wire Weld–Porosity

Small cavities or holes in the bead.

TOP
VIEW



POSSIBLE CAUSES AND SOLUTIONS

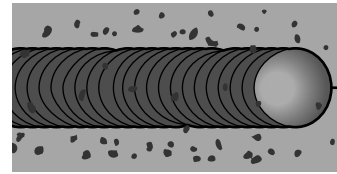
1. **Incorrect polarity:**
Check that polarity is set correctly for type of welding.
2. **Insufficient shielding gas (MIG only):**
Increase flow of gas.
Clean nozzle.
Maintain proper CTWD.
3. **Incorrect shielding gas (MIG only):**
Use shielding gas recommended by wire supplier.
4. **Dirty workpiece or welding wire:**
Clean workpiece down to bare metal.
Make certain that wire is clean and free from oil, coatings, and other residues.
5. **Inconsistent travel speed:**
Maintain steady travel speed.
6. **CTWD too long:**
Reduce CTWD.

Wire Weld–Excessive Spatter

Fine spatter is normal.

Spatter that is grainy and large is a problem.

TOP
VIEW

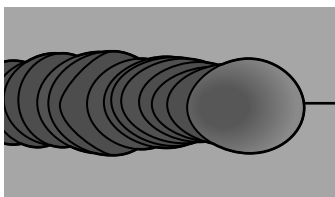


POSSIBLE CAUSES AND SOLUTIONS

1. **Dirty workpiece or welding wire:**
Clean workpiece down to bare metal.
Make certain that wire is clean and free from oil, coatings, and other residues.
2. **Incorrect polarity:**
Check that polarity is set correctly for type of welding.
3. **Insufficient shielding gas (MIG only):**
Increase flow of gas.
Clean nozzle.
Maintain proper CTWD.
4. **Wire feeding too fast:**
Reduce wire feed speed.
5. **CTWD too long:**
Reduce CTWD.

Wire Weld–Crooked/Wavy Bead

TOP
VIEW



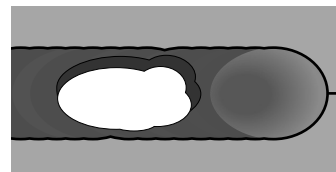
POSSIBLE CAUSES AND SOLUTIONS

1. **Inaccurate welding:**
Use two hands or rest hand on steady surface.
2. **Inconsistent travel speed:**
Maintain steady travel speed.
3. **CTWD too long:**
Reduce CTWD.

Wire Weld–Burn-Through

Base material melts away, leaving a hole in the weld.

TOP
VIEW



POSSIBLE CAUSES AND SOLUTIONS

1. **Workpiece overheating:**
Reduce current and/or wire feed speed.
2. **Travel speed too slow:**
Increase travel speed and ensure that travel speed is kept steady.
3. **Excessive material at weld:**
Reduce wire feed speed.

SAFETY

CONTROLS

WIRE

TIG / STICK

WELDING TIPS

MAINTENANCE